

0 Exercises

In Exercises 1–8 find the power series in x for the general solution.

1. $(1 + x^2)y'' + 6xy' + 6y = 0$
2. $(1 + x^2)y'' + 2xy' - 2y = 0$
3. $(1 + x^2)y'' - 8xy' + 20y = 0$
4. $(1 - x^2)y'' - 8xy' - 12y = 0$
5. $(1 + 2x^2)y'' + 7xy' + 2y = 0$
6. $(1 + x^2)y'' + 2xy' + \frac{1}{4}y = 0$
7. $(1 - x^2)y'' - 5xy' - 4y = 0$
8. $(1 + x^2)y'' - 10xy' + 28y = 0$

9. L

- (a) Find the power series in x for the general solution of $y'' + xy' + 2y = 0$.
 (b) For several choices of a_0 and a_1 , use differential equations software to solve the initial value problem

$$y'' + xy' + 2y = 0, \quad y(0) = a_0, \quad y'(0) = a_1, \quad (\text{A})$$

numerically on $(-5, 5)$.

- (c) For fixed r in $\{1, 2, 3, 4, 5\}$ graph

$$T_N(x) = \sum_{n=0}^N a_n x^n$$

and the solution obtained in (a) on $(-r, r)$. Continue increasing N until there's no perceptible difference between the two graphs.

10. L Follow the directions of Exercise 9 for the differential equation

$$y'' + 2xy' + 3y = 0.$$

In Exercises 11–13 find $a_0, \dots, a_N$ for N at least 7 in the power series solution $y = \sum_{n=0}^{\infty} a_n x^n$ of the initial value problem.

11. C $(1 + x^2)y'' + xy' + y = 0, \quad y(0) = 2, \quad y'(0) = -1$
12. C $(1 + 2x^2)y'' - 9xy' - 6y = 0, \quad y(0) = 1, \quad y'(0) = -1$
13. C $(1 + 8x^2)y'' + 2y = 0, \quad y(0) = 2, \quad y'(0) = -1$

14. L Do the next experiment for various choices of real numbers a_0, a_1 , and r , with $0 < r < 1/\sqrt{2}$.

- (a) Use differential equations software to solve the initial value problem

$$(1 - 2x^2)y'' - xy' + 3y = 0, \quad y(0) = a_0, \quad y'(0) = a_1, \quad (\text{A})$$

numerically on $(-r, r)$.

- (b) For $N = 2, 3, 4, \dots$, compute $a_2, \dots, a_N$ in the power series solution $y = \sum_{n=0}^{\infty} a_n x^n$ of (A), and graph

$$T_N(x) = \sum_{n=0}^N a_n x^n$$

and the solution obtained in (a) on $(-r, r)$. Continue increasing N until there's no perceptible difference between the two graphs.

15. L Do (a) and (b) for several values of r in $(0, 1)$:

- (a) Use differential equations software to solve the initial value problem

$$(1 + x^2)y'' + 10xy' + 14y = 0, \quad y(0) = 5, \quad y'(0) = 1, \quad (\text{A})$$

numerically on $(-r, r)$.

- (b) For $N = 2, 3, 4, \dots$, compute $a_2, \dots, a_N$ in the power series solution $y = \sum_{n=0}^{\infty} a_n x^n$ of (A), and graph

$$T_N(x) = \sum_{n=0}^N a_n x^n$$

and the solution obtained in (a) on $(-r, r)$. Continue increasing N until there's no perceptible difference between the two graphs. What happens to the required N as $r \rightarrow 1$?

- (c) Try (a) and (b) with $r = 1.2$. Explain your results.

In Exercises 16–20 find the power series in $x - x_0$ for the general solution.

16. $y'' - y = 0; \quad x_0 = 3$
17. $y'' - (x - 3)y' - y = 0; \quad x_0 = 3$
18. $(1 - 4x + 2x^2)y'' + 10(x - 1)y' + 6y = 0; \quad x_0 = 1$
19. $(11 - 8x + 2x^2)y'' - 16(x - 2)y' + 36y = 0; \quad x_0 = 2$
20. $(5 + 6x + 3x^2)y'' + 9(x + 1)y' + 3y = 0; \quad x_0 = -1$

In Exercises 21–26 find $a_0, \dots, a_N$ for N at least 7 in the power series $y = \sum_{n=0}^{\infty} a_n(x - x_0)^n$ for the solution of the initial value problem. Take x_0 to be the point where the initial conditions are imposed.

21. C $(x^2 - 4)y'' - xy' - 3y = 0, \quad y(0) = -1, \quad y'(0) = 2$
22. C $y'' + (x - 3)y' + 3y = 0, \quad y(3) = -2, \quad y'(3) = 3$
23. C $(5 - 6x + 3x^2)y'' + (x - 1)y' + 12y = 0, \quad y(1) = -1, \quad y'(1) = 1$
24. C $(4x^2 - 24x + 37)y'' + y = 0, \quad y(3) = 4, \quad y'(3) = -6$
25. C $(x^2 - 8x + 14)y'' - 8(x - 4)y' + 20y = 0, \quad y(4) = 3, \quad y'(4) = -4$
26. C $(2x^2 + 4x + 5)y'' - 20(x + 1)y' + 60y = 0, \quad y(-1) = 3, \quad y'(-1) = -3$
27. (a) Find a power series in x for the general solution of

$$(1 + x^2)y'' + 4xy' + 2y = 0. \quad (\text{A})$$

(b) Use (a) and the formula

$$\frac{1}{1 - r} = 1 + r + r^2 + \dots + r^n + \dots \quad (-1 < r < 1)$$

for the sum of a geometric series to find a closed form expression for the general solution of (A) on $(-1, 1)$.

- (c) Show that the expression obtained in (b) is actually the general solution of (A) on $(-\infty, \infty)$.
28. Use Theorem ?? to show that the power series in x for the general solution of

$$(1 + \alpha x^2)y'' + \beta xy' + \gamma y = 0$$

is

$$y = a_0 \sum_{m=0}^{\infty} (-1)^m \left[\prod_{j=0}^{m-1} p(2j) \right] \frac{x^{2m}}{(2m)!} + a_1 \sum_{m=0}^{\infty} (-1)^m \left[\prod_{j=0}^{m-1} p(2j+1) \right] \frac{x^{2m+1}}{(2m+1)!}.$$

29. Use Exercise 28 to show that all solutions of

$$(1 + \alpha x^2)y'' + \beta xy' + \gamma y = 0$$

are polynomials if and only if

$$\alpha n(n-1) + \beta n + \gamma = \alpha(n-2r)(n-2s-1),$$

where r and s are nonnegative integers.

30. (a) Use Exercise 28 to show that the power series in x for the general solution of

$$(1 - x^2)y'' - 2bxy' + \alpha(\alpha + 2b - 1)y = 0$$

is $y = a_0y_1 + a_1y_2$, where

$$y_1 = \sum_{m=0}^{\infty} \left[\prod_{j=0}^{m-1} (2j - \alpha)(2j + \alpha + 2b - 1) \right] \frac{x^{2m}}{(2m)!}$$

and

$$y_2 = \sum_{m=0}^{\infty} \left[\prod_{j=0}^{m-1} (2j + 1 - \alpha)(2j + \alpha + 2b) \right] \frac{x^{2m+1}}{(2m+1)!}.$$

- (b) Suppose $2b$ isn't a negative odd integer and k is a nonnegative integer. Show that y_1 is a polynomial of degree $2k$ such that $y_1(-x) = y_1(x)$ if $\alpha = 2k$, while y_2 is a polynomial of degree $2k + 1$ such that $y_2(-x) = -y_2(x)$ if $\alpha = 2k + 1$. Conclude that if n is a nonnegative integer, then there's a polynomial P_n of degree n such that $P_n(-x) = (-1)^n P_n(x)$ and

$$(1 - x^2)P_n'' - 2bP_n' + n(n + 2b - 1)P_n = 0. \quad (\text{A})$$

- (c) Show that (A) implies that

$$[(1 - x^2)^b P_n']' = -n(n + 2b - 1)(1 - x^2)^{b-1} P_n,$$

and use this to show that if m and n are nonnegative integers, then

$$\begin{aligned} [(1 - x^2)^b P_n']' P_m - [(1 - x^2)^b P_m']' P_n = \\ [m(m + 2b - 1) - n(n + 2b - 1)] (1 - x^2)^{b-1} P_m P_n. \end{aligned} \quad (\text{B})$$

- (d) Now suppose $b > 0$. Use (B) and integration by parts to show that if $m \neq n$, then

$$\int_{-1}^1 (1 - x^2)^{b-1} P_m(x) P_n(x) dx = 0.$$

(We say that P_m and P_n are *orthogonal on $(-1, 1)$ with respect to the weighting function $(1 - x^2)^{b-1}$* .)

31. (a) Use Exercise 28 to show that the power series in x for the general solution of Hermite's equation

$$y'' - 2xy' + 2\alpha y = 0$$

is $y = a_0y_1 + a_1y_2$, where

$$y_1 = \sum_{m=0}^{\infty} \left[\prod_{j=0}^{m-1} (2j - \alpha) \right] \frac{2^m x^{2m}}{(2m)!}$$

and

$$y_2 = \sum_{m=0}^{\infty} \left[\prod_{j=0}^{m-1} (2j + 1 - \alpha) \right] \frac{2^m x^{2m+1}}{(2m+1)!}.$$

- (b) Suppose k is a nonnegative integer. Show that y_1 is a polynomial of degree $2k$ such that $y_1(-x) = y_1(x)$ if $\alpha = 2k$, while y_2 is a polynomial of degree $2k + 1$ such that $y_2(-x) = -y_2(x)$ if $\alpha = 2k + 1$. Conclude that if n is a nonnegative integer then there's a polynomial P_n of degree n such that $P_n(-x) = (-1)^n P_n(x)$ and

$$P_n'' - 2xP_n' + 2nP_n = 0. \quad (\text{A})$$

- (c) Show that (A) implies that

$$[e^{-x^2} P_n']' = -2ne^{-x^2} P_n,$$

and use this to show that if m and n are nonnegative integers, then

$$[e^{-x^2} P_n']' P_m - [e^{-x^2} P_m']' P_n = 2(m - n)e^{-x^2} P_m P_n. \quad (\text{B})$$

- (d) Use (B) and integration by parts to show that if $m \neq n$, then

$$\int_{-\infty}^{\infty} e^{-x^2} P_m(x) P_n(x) dx = 0.$$

(We say that P_m and P_n are *orthogonal on* $(-\infty, \infty)$ *with respect to the weighting function* e^{-x^2} .)

- 32.** Consider the equation

$$(1 + \alpha x^3) y'' + \beta x^2 y' + \gamma xy = 0, \quad (\text{A})$$

and let $p(n) = \alpha n(n-1) + \beta n + \gamma$. (The special case $y'' - xy = 0$ of (A) is Airy's equation.)

- (a) Modify the argument used to prove Theorem ?? to show that

$$y = \sum_{n=0}^{\infty} a_n x^n$$

is a solution of (A) if and only if $a_2 = 0$ and

$$a_{n+3} = -\frac{p(n)}{(n+3)(n+2)} a_n, \quad n \geq 0.$$

- (b) Show from (a) that $a_n = 0$ unless $n = 3m$ or $n = 3m + 1$ for some nonnegative integer m , and that

$$a_{3m+3} = -\frac{p(3m)}{(3m+3)(3m+2)} a_{3m}, \quad m \geq 0,$$

and

$$a_{3m+4} = -\frac{p(3m+1)}{(3m+4)(3m+3)} a_{3m+1}, \quad m \geq 0,$$

where a_0 and a_1 may be specified arbitrarily.

(c) Conclude from (b) that the power series in x for the general solution of (A) is

$$y = a_0 \sum_{m=0}^{\infty} (-1)^m \left[\prod_{j=0}^{m-1} \frac{p(3j)}{3j+2} \right] \frac{x^{3m}}{3^m m!} \\ + a_1 \sum_{m=0}^{\infty} (-1)^m \left[\prod_{j=0}^{m-1} \frac{p(3j+1)}{3j+4} \right] \frac{x^{3m+1}}{3^m m!}.$$

In Exercises 33–37 use the method of Exercise 32 to find the power series in x for the general solution.

33. $y'' - xy = 0$

34. $(1 - 2x^3)y'' - 10x^2y' - 8xy = 0$

35. $(1 + x^3)y'' + 7x^2y' + 9xy = 0$

36. $(1 - 2x^3)y'' + 6x^2y' + 24xy = 0$

37. $(1 - x^3)y'' + 15x^2y' - 63xy = 0$

38. Consider the equation

$$(1 + \alpha x^{k+2})y'' + \beta x^{k+1}y' + \gamma x^k y = 0, \quad (\text{A})$$

where k is a positive integer, and let $p(n) = \alpha n(n-1) + \beta n + \gamma$.

(a) Modify the argument used to prove Theorem ?? to show that

$$y = \sum_{n=0}^{\infty} a_n x^n$$

is a solution of (A) if and only if $a_n = 0$ for $2 \leq n \leq k+1$ and

$$a_{n+k+2} = -\frac{p(n)}{(n+k+2)(n+k+1)} a_n, \quad n \geq 0.$$

(b) Show from (a) that $a_n = 0$ unless $n = (k+2)m$ or $n = (k+2)m+1$ for some nonnegative integer m , and that

$$a_{(k+2)(m+1)} = -\frac{p((k+2)m)}{(k+2)(m+1)[(k+2)(m+1)-1]} a_{(k+2)m}, \quad m \geq 0,$$

and

$$a_{(k+2)(m+1)+1} = -\frac{p((k+2)m+1)}{[(k+2)(m+1)+1](k+2)(m+1)} a_{(k+2)m+1}, \quad m \geq 0,$$

where a_0 and a_1 may be specified arbitrarily.

(c) Conclude from (b) that the power series in x for the general solution of (A) is

$$y = a_0 \sum_{m=0}^{\infty} (-1)^m \left[\prod_{j=0}^{m-1} \frac{p((k+2)j)}{(k+2)(j+1)-1} \right] \frac{x^{(k+2)m}}{(k+2)^m m!} \\ + a_1 \sum_{m=0}^{\infty} (-1)^m \left[\prod_{j=0}^{m-1} \frac{p((k+2)j+1)}{(k+2)(j+1)+1} \right] \frac{x^{(k+2)m+1}}{(k+2)^m m!}.$$

In Exercises 39–44 use the method of Exercise 38 to find the power series in x for the general solution.

39. $(1 + 2x^5)y'' + 14x^4y' + 10x^3y = 0$

40. $y'' + x^2y = 0$

41. $y'' + x^6y' + 7x^5y = 0$

42. $(1 + x^8)y'' - 16x^7y' + 72x^6y = 0$

43. $(1 - x^6)y'' - 12x^5y' - 30x^4y = 0$

44. $y'' + x^5y' + 6x^4y = 0$